

Assignment 1: Multivariate Statistics 2025/2026

Group 14

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Task 1

Question A

Performing PCA

All six variables were mean-centered and scaled before PCA was performed, so that the components are not dominated by differences in the scales between the variables.

```
> life_standard <- scale(life, center = TRUE, scale = TRUE)
> life_pca <- prcomp(life_standard)
> summary(life_pca)
Importance of components:

            PC1      PC2      PC3      PC4      PC5      PC6
Standard deviation  1.5491  1.2605  0.9486  0.70591  0.61983  0.47857
Proportion of Variance 0.3999  0.2648  0.1500  0.08305  0.06403  0.03817
Cumulative Proportion 0.3999  0.6648  0.8147  0.89780  0.96183  1.00000
```

The first two components explain 66,5% of the variance, adding PC3 to the model brings this to 81,5% of the variance explained. However, only the first 2 components seem to have eigenvalues greater than 1, indicating that only they explain more variance than any one of the variables on their own.

```
> loadings <- life_pca$rotation %*% diag(life_pca$sdev)
> round(loadings, 3)

      [,1] [,2] [,3] [,4] [,5] [,6]
family    0.733 -0.374 -0.156 -0.480 -0.235 -0.112
friends   -0.335 -0.816  0.123 -0.189  0.414 -0.004
leisure_time -0.477 -0.249 -0.827  0.025 -0.065  0.146
politics    0.151 -0.847  0.169  0.402 -0.264 -0.019
work        0.811  0.052 -0.381  0.263  0.267 -0.234
religion    0.918 -0.034  0.049  0.029  0.113  0.374
```

To improve interpretability, we can also invert PC2 and PC3 so that all three components have positive loadings. Flipping the scores and then recalculating the component scores and loadings produces the following results:

- **Traditional priority (PC1):** post-inversion, this principal component has strong positive loadings on **religion** (0.918), **work** (0.811), and **family** (0.733). The other variables have relatively weaker loadings. A high score on PC1 corresponds with people valuing **religion**, **family**, and **work** highly, whereas a low score on PC1 shows weaker importance on these more traditional values
- **Social-civic priority (PC2):** post-inversion, the component has large loadings on **friends** (0.816) and **politics** (0.847), while the other variables have relatively weaker loadings. A high score on PC2 indicates importance given to **friendships** and **political awareness**. A low score on PC2 indicates a more individualistic (less reliance on social groups and peers) and apolitical (less importance for politics) worldview

- **Leisure time (PC3):** post-inversion, this component is dominated almost exclusively by a strong loading on `leisure time` (0.827). This suggests a mainly single-item component, where a high score on PC3 shows that more importance is placed on leisure and a low score indicates the opposite

Horn's Procedure

```
screeplot(life_pca, type = "lines", main = "Scree plot",
          xlim = c(0.75, 6.25), ylim = c(0, 3))
```

```
# comparing against the mean
```

```
set.seed(17)
```

```
paran(
```

```
  life_standard,
```

```
  iterations = 5000,
```

```
  graph = TRUE,
```

```
  cfa = FALSE,
```

```
  centile = 0
```

```
)
```

```
# comparing against the 95th percentile
```

```
set.seed(21)
```

```
paran(
```

```
  life_standard,
```

```
  iterations = 5000,
```

```
  graph = TRUE,
```

```
  cfa = FALSE,
```

```
  centile = 95
```

```
)
```

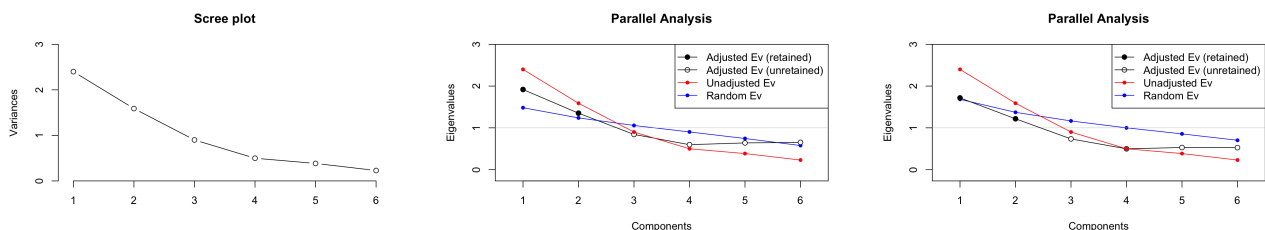


Figure 1: Scree plot and Horn's procedure (respectively for the mean and the 95th percentile) results.

The scree-plot, the rule of thumb of eigenvalue > 1 , as well as Horn's parallel analysis would suggest a two-component model. Keeping only two components would also make sense conceptually, as PC3 essentially represents a single-variable component capturing the importance people place on `leisure time`, it would be clearer to include `leisure time` directly in the model rather than adding a third component that mainly reflects this variable in a less transparent way.

Question B

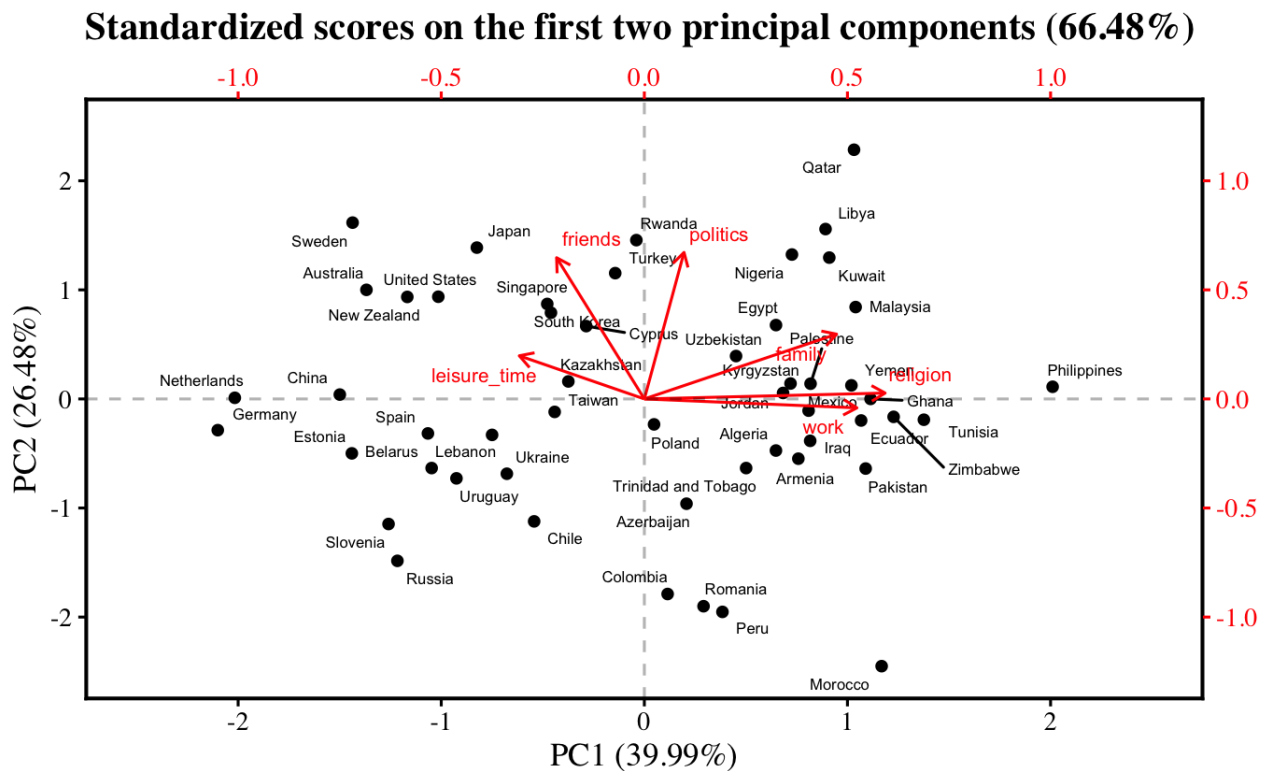


Figure 2: Biplot for the first two CP

The biplot in figure 2 allows us to categorize countries based on the quadrant of the biplot they fall into:

- **Top-right quadrant (high PC1 and high PC2):** countries that score high on both components tend to value **family**, **work**, **religion**, **friends** and **politics**. They remain traditional and duty-focused at the same time, while also not shying away from social obligations towards peers. Examples of countries in this quadrant are countries such as Ghana, Yemen, and the Philippines.
- **Top-left quadrant (low PC1 and high PC2):** countries in this quadrant value **friends** and **politics** more than their work and family, and tend to be more progressive countries that emphasize individuality and awareness, such as the US, Sweden, and Singapore.
- **Bottom-left quadrant (low PC1 and low PC2):** countries with low scores on both components tend to be quite neutral and don't place undue importance on any one variable, such as Germany, the Netherlands, and Slovenia.
- **Bottom-right quadrant (high PC1 and low PC2):** countries in the bottom right quadrant value a traditional outlook and give more importance to **family**, **work**, and **religion** over **friends** and **politics**, such as Romania, Pakistan, and Zimbabwe.

The biplot also shows that **family**, **work** and **religion** point roughly in the same direction along PC1, which tends to suggest that usually countries that score high on one variable would also score higher on the other. Meanwhile, **friends** and **politics** point strongly along PC2 but are only moderately correlated, as highlighted by the slightly larger angle between the arrows representing them, when compared to the more acute angles between the **work** and **religion** arrows. **family** and **friends** form an angle of approximately 90°, which means they are close to be uncorrelated. **leisure_time** is somewhat stray since this is the variable that is least represented by the chosen components. It is also noticeable that it has the shortest length.

Task 2

Question A

Exploratory Factor Analysis

All items were mean-centered and scaled so that factor extraction/rotation would not be affected by differences in scale. The results of the factor analysis with 5 factors are shown below:

```
> df_standard <- as.data.frame(scale(fldata[, 2:ncol], center =  
+                                T, scale = T))  
> cor_df <- cor(df_standard)  
> efa_df <- fa(  
+   r = cor_df,  
+   fm = "ml",  
+   nfactors = 5,  
+   rotate = "oblimin",  
+   scores = "regression",  
+   n.obs = nrow  
+ )  
> efa_df$loadings
```

Loadings:

	ML1	ML2	ML3	ML5	ML4
FS_pay_bills		0.563			-0.196
FS_afford_extras		0.868			
FS_afford_housing		0.569			-0.158
FS_save_money		0.766			
FSF_pay_bills	0.692				-0.155
FSF_afford_extras	0.856				
FSF_afford_housing	0.679				-0.217
FSF_save_money	0.852				0.113
SFJ_no_info				0.480	0.154
SFJ_no_chance_show				0.677	
SFJ_no_training				0.640	
SFJ_no_support_findjob				0.556	
SDJ_help_people			0.615		0.115
SDJ_learn_new_things			0.629		-0.121
SDJ_develop_creativity			0.585		
SDJ_meet_people			0.649		
SDJ_feeling_self_worth			0.541		-0.182
HEALTH_felt_down					0.712
HEALTH_limitation					0.744

	ML1	ML2	ML3	ML5	ML4
SS loadings	2.405	1.994	1.834	1.433	1.320
Proportion Var	0.127	0.105	0.097	0.075	0.069

```

Cumulative Var 0.127 0.232 0.328 0.403 0.473
> data.frame(communality = round(efa_df$communalities, 3),
+            uniquenesses = round(efa_df$uniquenesses, 3))
      communality uniquenesses
FS_pay_bills      0.369      0.631
FS_afford_extras  0.754      0.246
FS_afford_housing  0.405      0.595
FS_save_money     0.578      0.422
FSF_pay_bills     0.514      0.486
FSF_afford_extras 0.740      0.260
FSF_afford_housing 0.546      0.454
FSF_save_money    0.719      0.281
SFJ_no_info       0.284      0.716
SFJ_no_chance_show 0.451      0.549
SFJ_no_training   0.428      0.572
SFJ_no_support_findjob 0.330      0.670
SDJ_help_people   0.398      0.602
SDJ_learn_new_things 0.419      0.581
SDJ_develop_creativity 0.366      0.634
SDJ_meet_people   0.407      0.593
SDJ_feeling_self_worth 0.336      0.664
HEALTH_felt_down  0.562      0.438
HEALTH_limitation 0.554      0.446

```

All 5 factors together explain only 47.3% of the total variance, with ML1+ML2 explaining 23.2% of the variance within the data. Moreover, we can say that:

- ML1 provides a clean look at the person's parents' financial situation, and there is good separation as all 4 FSF items have high loadings on this factor, while no other variables do.
- ML2 explains the current financial situation and has similar loading to ML1, but on the FS variables. Similar loadings also make sense conceptually, as both ML1 and ML2 deal with theoretically similar variables, which describe the past and present financial situation of an individual.
- ML3 has high loadings on SDJ components, which deal with self-development through jobs, but the loadings themselves are less strong than the ones observed for ML1 and ML2
- ML4 deals with the person's health and has high loadings on both variables associated with health.
- ML5 loads highly on SFJ or support vectors, but just as with SDJ, the loadings are less strong than ML1/ML2 .

Within ML1, `FS_afford_extras` and `save_money` are captured well, but even the other variables are moderately represented by ML1. Within ML2, all FSF variables are well represented and have communality > 0.5 , which means that ML2 is a good explanatory latent variable. SFJ and SDJ are represented worse with `SFJ_no_info` and `SDJ_develop_creativity` being the least represented. Both health variables have communality > 0.5 , showing they are well-explained.

Reproducibility

```

> rep_cor <- efa_df$model
> change <- max(abs(cor_df - rep_cor))
> rms_res <- sqrt(mean((cor_df - rep_cor)[lower.tri(cor_df)]^2))
> c(change, rms_res)
[1] 0.71626483 0.02213703
> efa_df$STATISTIC
[1] 1417.338
> efa_df$PVAL

```

[1] 6.706579e-240

The largest absolute difference between the observed and reproduced correlation matrices is 0.716, which indicates that some of the variables are not well-reproduced, leading to larger residuals. However, the *RMS* is close to zero, which means close-to-perfect average reproduction of correlations when the whole dataset is considered. Since on average the reproducibility is good, it might be that just one or two variables are not being reproduced effectively. The χ^2 amounts to 1417.338 with a *p*-value $< .001$, so we may conclude that the model does not perfectly fit the data. However, since we know that for large sample sizes, χ^2 is sensitive to relatively small misfits, we must use other fit measures to evaluate the model and rely on other tests to confirm our factor selections.

Question B

Confirmatory Factor Analysis

After conducting a CFA for a model with 5 correlated latent variables (one for each concept), where each item only has a loading on the concept it aims to measure, the fit measures indices we get are:

```
> cfa_model_1 <- '
+ FS=~NA*FS_pay_bills+FS_afford_extras+FS_afford_housing+FS_save_money
+ FSF=~NA*FSF_pay_bills+FSF_afford_extras+FSF_afford_housing+FSF_save_money
+ SFJ=~NA*SFJ_no_info+SFJ_no_chance_show+SFJ_no_training+SFJ_no_support_findjob
+ SDJ=~NA*SDJ_help_people+SDJ_learn_new_things+SDJ_develop_creativity+
+ SDJ_meet_people+SDJ_feeling_self_worth
+ HEALTH=~NA*HEALTH_felt_down+HEALTH_limitation
+ FS ~~ 1*FS
+ FSF ~~ 1*FSF
+ SFJ ~~ 1*SFJ
+ SDJ ~~ 1*SDJ
+ HEALTH ~~ 1*HEALTH
+ FS ~~ FSF
+ FS ~~ SFJ
+ FS ~~ SDJ
+ FS ~~ HEALTH
+ FSF ~~ SFJ
+ FSF ~~ SDJ
+ FSF ~~ HEALTH
+ SFJ ~~ SDJ
+ SFJ ~~ HEALTH
+ SDJ ~~ HEALTH
+ '
> cfa_fit_1 <- cfa(model = cfa_model_1,
+                 sample.cov = cov_df,
+                 sample.nobs = sample_size)
> fitmeasures(cfa_fit_1, c("chisq", "df", "pvalue", "cfi", "tli", "rmsea", "SRMR"))
  chisq    df  pvalue   cfi   tli  rmsea  srmr
2785.507 142.000   0.000  0.906  0.886  0.062  0.044
```

The high *chisq* and low *p*-value may indicate that the model significantly differs from the data, but since we are working with a large sample size (4829), we should base our analysis on the other parameters. *CFI* is above 0.90 and while *TLI* is less than 0.90, it is marginally so. Both *RMSEA* and *SRMR* are below 0.08 and show a decent fit, albeit not a great one.

```
> summary(cfa_fit_1)$pe[, c(1:3, 5)]
      lhs op      rhs      est
```

1	FS	=~	FS_pay_bills	0.445844832
2	FS	=~	FS_afford_extras	0.848930064
3	FS	=~	FS_afford_housing	0.675233313
4	FS	=~	FS_save_money	0.827721452
5	FSF	=~	FSF_pay_bills	0.443230244
6	FSF	=~	FSF_afford_extras	0.840667601
7	FSF	=~	FSF_afford_housing	0.573497015
8	FSF	=~	FSF_save_money	0.912859227
9	SFJ	=~	SFJ_no_info	0.427004796
10	SFJ	=~	SFJ_no_chance_show	0.499155894
11	SFJ	=~	SFJ_no_training	0.536521242
12	SFJ	=~	SFJ_no_support_findjob	0.469561461
13	SDJ	=~	SDJ_help_people	0.497943967
14	SDJ	=~	SDJ_learn_new_things	0.424779438
15	SDJ	=~	SDJ_develop_creativity	0.484346281
16	SDJ	=~	SDJ_meet_people	0.515093096
17	SDJ	=~	SDJ_feeling_self_worth	0.348166629
18	HEALTH	=~	HEALTH_felt_down	0.832086616
19	HEALTH	=~	HEALTH_limitation	0.551849782
20	FS	~~	FS	1.000000000
21	FSF	~~	FSF	1.000000000
22	SFJ	~~	SFJ	1.000000000
23	SDJ	~~	SDJ	1.000000000
24	HEALTH	~~	HEALTH	1.000000000
25	FS	~~	FSF	0.331787494
26	FS	~~	SFJ	-0.236547950
27	FS	~~	SDJ	-0.004331225
28	FS	~~	HEALTH	-0.259732829
29	FSF	~~	SFJ	-0.092885957
30	FSF	~~	SDJ	0.042523159
31	FSF	~~	HEALTH	-0.142294883
32	SFJ	~~	SDJ	0.317931353
33	SFJ	~~	HEALTH	0.260430959
34	SDJ	~~	HEALTH	0.048936818
35	FS_pay_bills	~~	FS_pay_bills	0.386193363
36	FS_afford_extras	~~	FS_afford_extras	0.254160332
37	FS_afford_housing	~~	FS_afford_housing	0.707890564
38	FS_save_money	~~	FS_save_money	0.542501537
39	FSF_pay_bills	~~	FSF_pay_bills	0.207722624
40	FSF_afford_extras	~~	FSF_afford_extras	0.248523593
41	FSF_afford_housing	~~	FSF_afford_housing	0.341018851
42	FSF_save_money	~~	FSF_save_money	0.365626464
43	SFJ_no_info	~~	SFJ_no_info	0.477088313
44	SFJ_no_chance_show	~~	SFJ_no_chance_show	0.364770573
45	SFJ_no_training	~~	SFJ_no_training	0.362373220
46	SFJ_no_support_findjob	~~	SFJ_no_support_findjob	0.434001286
47	SDJ_help_people	~~	SDJ_help_people	0.425412826
48	SDJ_learn_new_things	~~	SDJ_learn_new_things	0.255921846
49	SDJ_develop_creativity	~~	SDJ_develop_creativity	0.404316102
50	SDJ_meet_people	~~	SDJ_meet_people	0.437547902
51	SDJ_feeling_self_worth	~~	SDJ_feeling_self_worth	0.276540207
52	HEALTH_felt_down	~~	HEALTH_felt_down	0.067054549
53	HEALTH_limitation	~~	HEALTH_limitation	0.506050384

Convergence

```
> ave <- function(load){
+   load <- load[load != 0]
+   sum(load^2) / length(load)
+ }
> AVE_res <- apply(std_loadings, 2, ave)
> round(AVE_res, 3)
      FS      FSF      SFJ      SDJ HEALTH
0.507  0.603  0.365  0.366  0.644
```

Convergent validity tells us whether items that measure the same construct are statistically related. As we can see from the *AVE* calculation, FS, FSF, and HEALTH have good convergent validity since they possess an $AVE \geq 0.50$ with SFJ and SDJ showing weaker convergent validity with $AVE < 0.50$.

Divergence

Discriminant validity is supported, as none of the correlations between latent factors is as high as the variance explained within each factor (as indicated by the higher *AVE* values). All inter-factor correlations remain well below 0.85, further confirming discriminant validity.

Composite Reliability

When calculating the composite reliability, we ignore the occurrences where the std loadings is 0 since the square matrix returned by `lavaan` also lists the indicators that do not load on a factor. As a consequence, we only take the indicators that load on it within the model.

```
> comp_rel <- function(loads, errs)
+ {
+   (sum(loads)^2) / ((sum(loads)^2) + sum(errs))
+ }
> composite_results <- (1:ncol(std_loadings), function(i) {
+   l <- std_loadings[, i][std_loadings[, i] != 0]
+   e <- err_var[, i][std_loadings[, i] != 0]
+   comp_rel(l, e)
+ })
> composite_results
[1] 0.8008874 0.8575254 0.6956659 0.7424298 0.7752427
```

The set of observed indicators exhibits a high degree of internal consistency in measuring each latent construct, reflecting the reliability and stability of the construct representation in the model.

Question C

Modification Indices

```
> mi_fit1_sorted[1:20, 1:4]
      lhs op      rhs      mi
211  FSF_afford_extras ~~      FSF_save_money 578.039
197   FSF_pay_bills ~~      FSF_afford_housing 520.574
282   SDJ_help_people ~~      SDJ_meet_people 356.694
149   FS_afford_extras ~~      FS_save_money 248.208
168   FS_afford_housing ~~      FSF_afford_housing 199.372
133    FS_pay_bills ~~      FSF_pay_bills 191.763
119      HEALTH =~      FSF_afford_housing 189.541
131    FS_pay_bills ~~      FS_afford_housing 140.610
```

184	FS_save_money	~~	FSF_save_money	135.216
210	FSF_afford_extras	~~	FSF_afford_housing	133.178
223	FSF_afford_housing	~~	FSF_save_money	110.506
196	FSF_pay_bills	~~	FSF_afford_extras	108.231
151	FS_afford_extras	~~	FSF_afford_extras	103.183
120	HEALTH	=~	FSF_save_money	98.243
288	SDJ_learn_new_things	~~	SDJ_feeling_self_worth	98.062
286	SDJ_learn_new_things	~~	SDJ_develop_creativity	95.488
198	FSF_pay_bills	~~	FSF_save_money	94.140
287	SDJ_learn_new_things	~~	SDJ_meet_people	87.557
280	SDJ_help_people	~~	SDJ_learn_new_things	86.576
125	HEALTH	=~	SDJ_help_people	81.776

Several very high *MI* pairs due to the highly similar factors as seen earlier. As suspected, several FSF and FS terms have high correlation, which makes sense conceptually. For example, people who think they can afford extra things would also believe they could save money. Picking variables within the same grouping/correlated groups and which makes sense conceptually for now, to test whether those would give a better score.

Alternative CFA Model

In the new model, we add the following correlations:

- FSF_afford_extras ~~ FSF_save_money
- SDJ_help_people ~~ SDJ_meet_people
- SDJ_learn_new_things ~~ SDJ_develop_creativity
- FS_pay_bills ~~ FSF_pay_bill
- FS_afford_housing ~~ FSF_afford_housing
- FS_save_money ~~ FSF_save_money
- FS_afford_extras ~~ FSF_afford_extras
- FS_afford_extras ~~ FS_save_money

```
> cfa_model_2 <- '
+ FS=~NA*FS_pay_bills+FS_afford_extras+FS_afford_housing+FS_save_money
+ FSF=~NA*FSF_pay_bills+FSF_afford_extras+FSF_afford_housing+FSF_save_money
+ SFJ=~NA*SFJ_no_info+SFJ_no_chance_show+SFJ_no_training+SFJ_no_support_findjob
+ SDJ=~NA*SDJ_help_people+SDJ_learn_new_things+SDJ_develop_creativity+
+ SDJ_meet_people+SDJ_feeling_self_worth
+ HEALTH=~NA*HEALTH_felt_down+HEALTH_limitation
+ FS ~~ 1*FS
+ FSF ~~ 1*FSF
+ SFJ ~~ 1*SFJ
+ SDJ ~~ 1*SDJ
+ HEALTH ~~ 1*HEALTH
+ FS ~~ FSF
+ FS ~~ SFJ
+ FS ~~ SDJ
+ FS ~~ HEALTH
+ FSF ~~ SFJ
+ FSF ~~ SDJ
+ FSF ~~ HEALTH
+ SFJ ~~ SDJ
+ SFJ ~~ HEALTH
+ SDJ ~~ HEALTH
+ FSF_afford_extras ~~ FSF_save_money
+ SDJ_help_people ~~ SDJ_meet_people
```

```

+ SDJ_learn_new_things ~~ SDJ_develop_creativity
+ FS_pay_bills ~~ FSF_pay_bills
+ FS_afford_housing ~~ FSF_afford_housing
+ FS_save_money ~~ FSF_save_money
+ FS_afford_extras ~~ FSF_afford_extras
+ FS_save_money ~~ FS_afford_extras
+ '
> cfa_fit_2 <- cfa(model = cfa_model_2,
+                 sample.cov = cov_df,
+                 sample.nobs = sample_size)
> fitmeasures(cfa_fit_2, c("chisq", "df", "pvalue", "cfi", "tli", "rmsea", "SRMR"))
      chisq      df    pvalue      cfi      tli    rmsea    srmr
1175.227  134.000    0.000    0.963    0.953    0.040    0.035

```

After including the suggested modification indices to include the correlated error terms, we get the following fit indices:

- χ^2 has dropped significantly to 1175.227 but is still significant with a p -value < 0.001 , indicating that the model is not a perfect fit. However, as discussed earlier, this may be due to the large sample size, and so we need to use the other fitness metrics to judge the model. Overall, it is safe to say that the significant reduction from 2785 to 1175 indicates that adding the correlated errors has improved our model.
- Both *CFI* and *TLI* are now greater than 0.95 meaning excellent comparative fit.
- *RMSEA* less than 0.040, which shows a 0.02 drop, showing good model improvement. The *SRMR* of 0.035 also gives credence to an excellent fit.
- Information criteria AIC and BIC have respectively dropped from 203341 to 201747.208 and 203653 to 202110.222, which is further evidence of the fact that we improved our fit.

Interpretation of Correlated Error Terms

- *FSF_afford_extras* $\sim$ *FSF_save_money*: both relate to financial management within the family, and it makes sense that errors would be correlated because those who save money can afford to buy extras. Both concern decisions regarding how to handle extra money.
- *SDJ_help_people* $\sim$ *SDJ_meet_people*: both relate to behaviours that dictate an individual's social interactions, and it makes conceptual sense that those who help people tend to be social and look forward to meeting new people and engaging with them. The residual correlation shows that these attributes share some social behaviour that may not fully be captured by the SDJ factor.
- *SDJ_learn_new_things* $\sim$ *SDJ_develop_creativity*: both relate to how an individual handles personal growth or self-development. Folks who are creative tend to also enjoy new things and vice versa. Cognitive functioning may be supported or enhanced by being creative, and the residual correlation might be capturing this overlap.
- *FS_pay_bills* $\sim$ *FSF_pay_bills*: both these variables measure the ability to pay bills, but in different contexts, which may be related since one relates to personal capacity and the other to the family's capacity. Since this capacity might have some temporal context, the residual correlation makes conceptual sense.
- *FS_afford_housing* $\sim$ *FSF_afford_housing*: both measure the ability to afford housing in a personal against family context, and the correlation may be capturing this overlap
- *FS_save_money* $\sim$ *FSF_save_money*: both talk about the ability to save money and propensity to do so, and there may be overlap between these decisions regarding personal finance for an individual vs their family, which has been captured.
- *FS_afford_extras* $\sim$ *FSF_afford_extras*: both measure a person's and family's ability to afford discretionary spending, or purchase additional items which are not a necessity.
- *FS_afford_extras* $\sim$ *FS_save_money*: these are within the same factor and both contribute information regarding choices concerning personal financial planning. Since they both capture a similar behavioural pattern, it makes sense that it would have residual correlation.

Question D

SEM

We estimated four versions of the multi-group structural equation model:

1. configural invariance model
2. configural invariance model with equal regression coefficients: the additional equation is

$$FS \sim \beta_1 FSF + \beta_2 SFJ + \beta_3 SDJ + \beta_4 HEALTH$$

3. metric invariance model with country-specific regression: same model as 1, but with additional constraints for the loadings across groups
4. metric invariance model with equal regression coefficients: same model as 2, but with additional constraints for the loadings across groups

```
config_invariance <- sem(  
  model_structural_free,  
  data = df_centered,  
  group = "country"  
)  
config_invariance_equal <- sem(  
  model_structural_constrained,  
  data = df_centered,  
  group = "country",  
  group.equal = "regressions"  
)  
metric_invariance <- sem(  
  model_structural_free,  
  data = df_centered,  
  group = "country",  
  group.equal = "loadings"  
)  
metric_invariance_equal <- sem(  
  model_structural_constrained,  
  data = df_centered,  
  group = "country",  
  group.equal = c("loadings", "regressions")  
)
```

Fit Comparison

As per fit indices, we can observe the following:

```
> fitmeasures(config_invariance,  
+             c("chisq", "df", "cfi", "tli", "rmsea", "srmr", "aic", "bic"))  
      chisq      df      cfi      tli      rmsea      srmr      aic  
1394.110    268.000    0.960    0.949    0.042    0.036 200969.453  
      bic  
201941.813  
> fitmeasures(config_invariance_equal,  
+             c("chisq", "df", "cfi", "tli", "rmsea", "srmr", "aic", "bic"))  
      chisq      df      cfi      tli      rmsea      srmr      aic  
1397.264    272.000    0.960    0.950    0.041    0.036 200964.606  
      bic  
201911.036  
> fitmeasures(metric_invariance,
```

```

+           c("chisq", "df", "cfi", "tli", "rmsea", "srmr", "aic", "bic"))
      chisq      df      cfi      tli      rmsea      srmr      aic
1536.241    287.000    0.955    0.947    0.042    0.045 201073.584
      bic
201922.778
> fitmeasures(metric_invariance_equal,
+           c("chisq", "df", "cfi", "tli", "rmsea", "srmr", "aic", "bic"))
      chisq      df      cfi      tli      rmsea      srmr      aic
1538.925    291.000    0.956    0.948    0.042    0.046 201068.268
      bic
201891.532

```

As per the fit indices we can observe the following:

- All have excellent *CFI* and *TLI* number $\geq .95$, and *RMSEA* and *SRMR* $< .05$ indicating that every model fits the data decently.
- Compare *CFI* and *RMSEA* δ 's of the free and equal regression models to see if it makes sense to assume the relationships between variables to be same across countries shows that the drop is negligible (configural level - delta minimal (0.0003), metric level - delta minimal (0.0003)).

Hence, constraining the regression coefficients does not worsen model fit meaning that the direction and strength of the relationship is equivalent across countries which supports structural invariance. The configural with fixed regression also gives us more degrees of freedom than with free regression, while providing better fit measures and does not impose as many restrictions as the metric invariant regression constrained model making it more parsimonious while also aiding interpretability by enabling comparison of latent variable relationships across countries.

Owing to having chosen the configural model with regression coefficients constrained, we cannot infer information regarding differences in group means from the model intercepts since the loadings for the countries have not been constrained.

```

> standard_solution <- standardizedSolution(config_invariance_equal)
> standard_regressions <- subset(standard_solution, op == "~")
> standard_regressions

```

	lhs	op	rhs	group	label	est.std	se	z	pvalue	ci.lower	ci.upper
39	FS	~	FSF	1	a1	0.256	0.017	14.693	0.000	0.222	0.290
40	FS	~	SFJ	1	a2	-0.171	0.021	-8.027	0.000	-0.212	-0.129
41	FS	~	SDJ	1	a3	0.055	0.020	2.706	0.007	0.015	0.094
42	FS	~	HEALTH	1	a4	-0.220	0.019	-11.654	0.000	-0.257	-0.183
124	FS	~	FSF	2	a1	0.257	0.018	14.670	0.000	0.222	0.291
125	FS	~	SFJ	2	a2	-0.171	0.021	-8.016	0.000	-0.213	-0.129
126	FS	~	SDJ	2	a3	0.055	0.020	2.712	0.007	0.015	0.094
127	FS	~	HEALTH	2	a4	-0.221	0.019	-11.664	0.000	-0.258	-0.184

Since we have opted for the covariance model with equal regression coefficients we cannot compare between groups, as the coefficients would be multiplied by variables which are scaled differently due to different loadings and residuals. However, we can infer about the strength of the effect that each latent variable has on predicting FS. It can be inferred that holding other variables constant a unit change in FSF has the most effect on FS, when compared with SFJ, SDJ and Health. Also, both the Health and SFJ seem to have negative relationships with FS as compared to SDJ and FSF.

If the need required comparison between groups, we could opt for the metric invariance model with regression coefficients constrained so that the variables and their impact could be compared between countries after validating using supplementary information that could validate that the variables load similarly in these two countries.